

A Cash Flow-Fingerprint Model for Quantitative ETF Sector Rotation Analysis

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Research Focus

Research Question: Can a cash flow-dominant multi-factor model outperform traditional momentum-based sector rotation strategies?

Core Hypothesis: Assigning cash flow factors a weight >30% generates more stable and profitable signals.

Research Value: Provide retail investors with a transparent, interpretable quantitative investment tool.

Cash Flow Influence



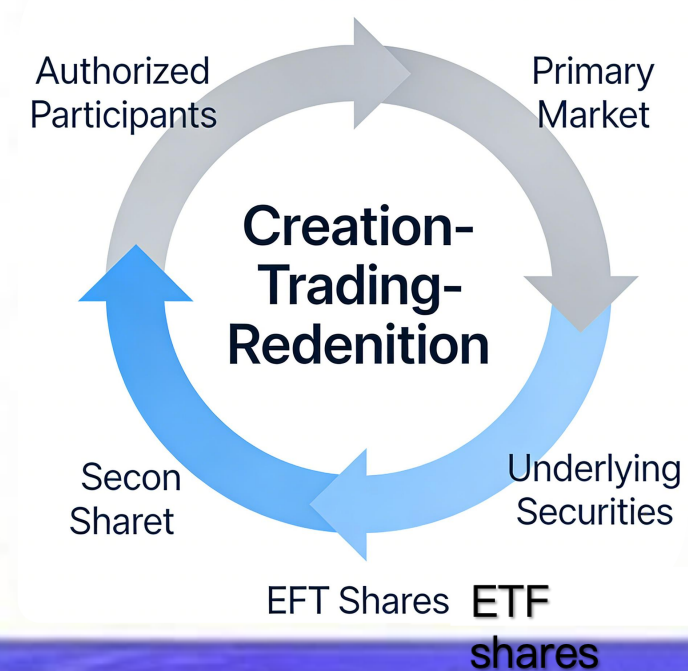
Originality

Existing Limitations: Traditional models overlook cash flow signals; ML models lack transparency.

Innovations:

- ① First to enforce cash flow weight constraint (35%);
- ② Integrate financial theory with engineering implementation;
- ③ Noise filtering for time series.

Traditional ETF Model



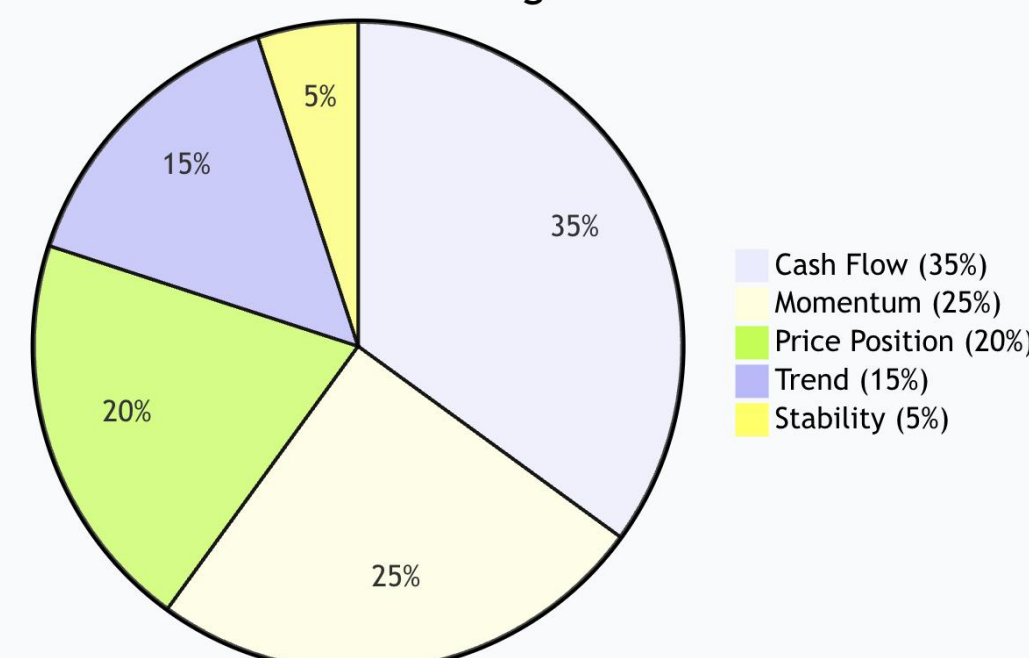
Methodology

Data: 15 Chinese sector ETFs (2020-2024), Baostock/AkShare APIs

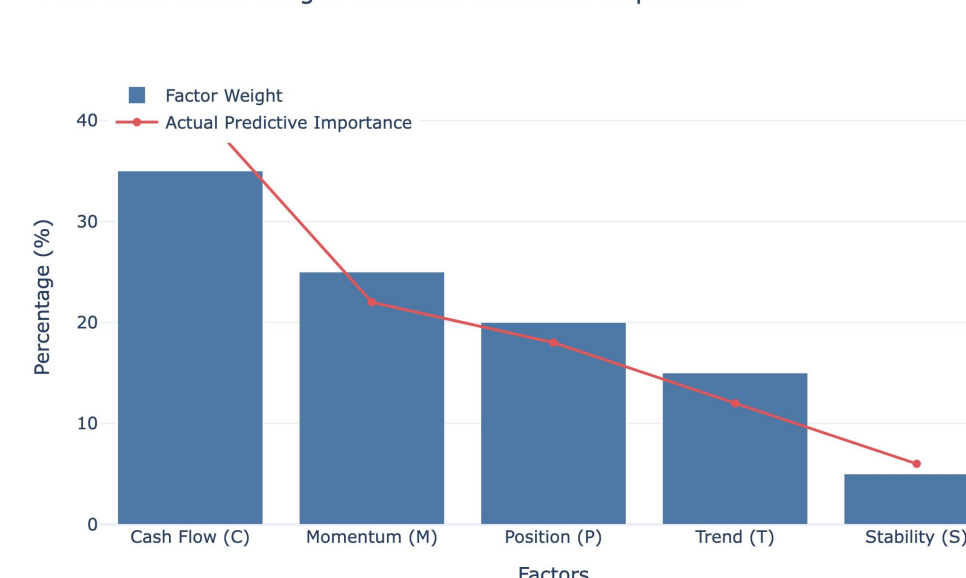
Factor System (FlowPrint Score): Cash Flow (35%) + Momentum (25%) + Price Position (20%) + Trend Strength (15%) + Stability (5%)

Workflow: Factor Calculation → Normalization → Weighted Scoring → Weekly Ranking → Backtest Validation

FlowPrint Factor Weights



FlowPrint: Factor Weight vs. Actual Predictive Importance

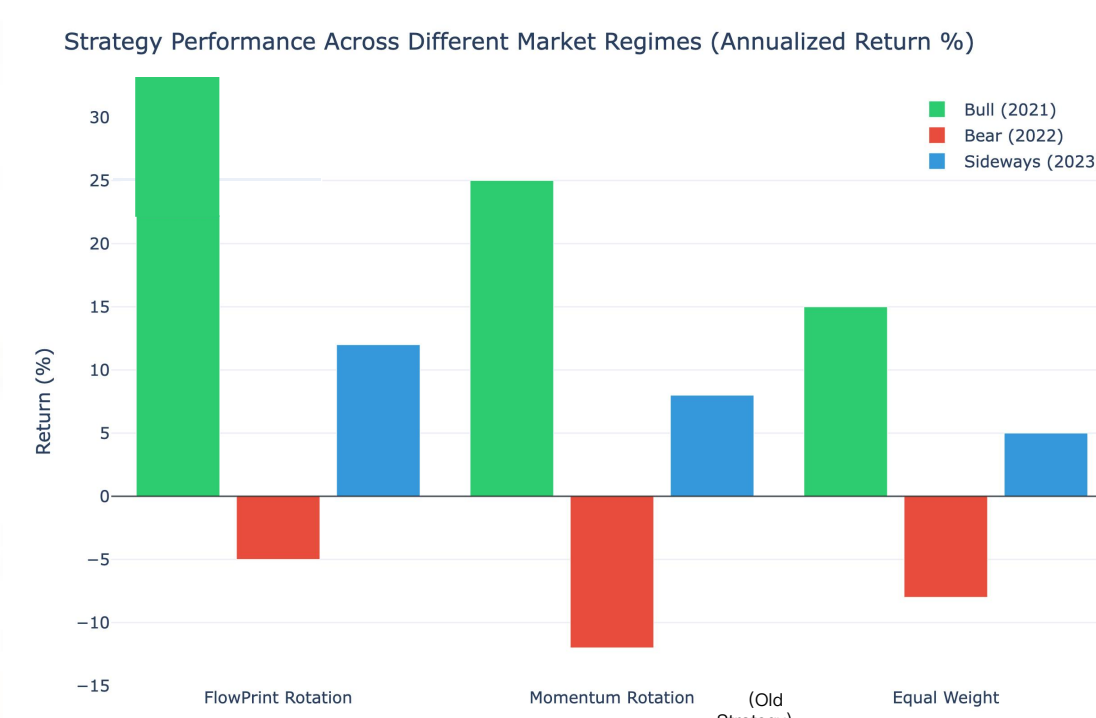


Key Results (2023-2024 Out-of-Sample)

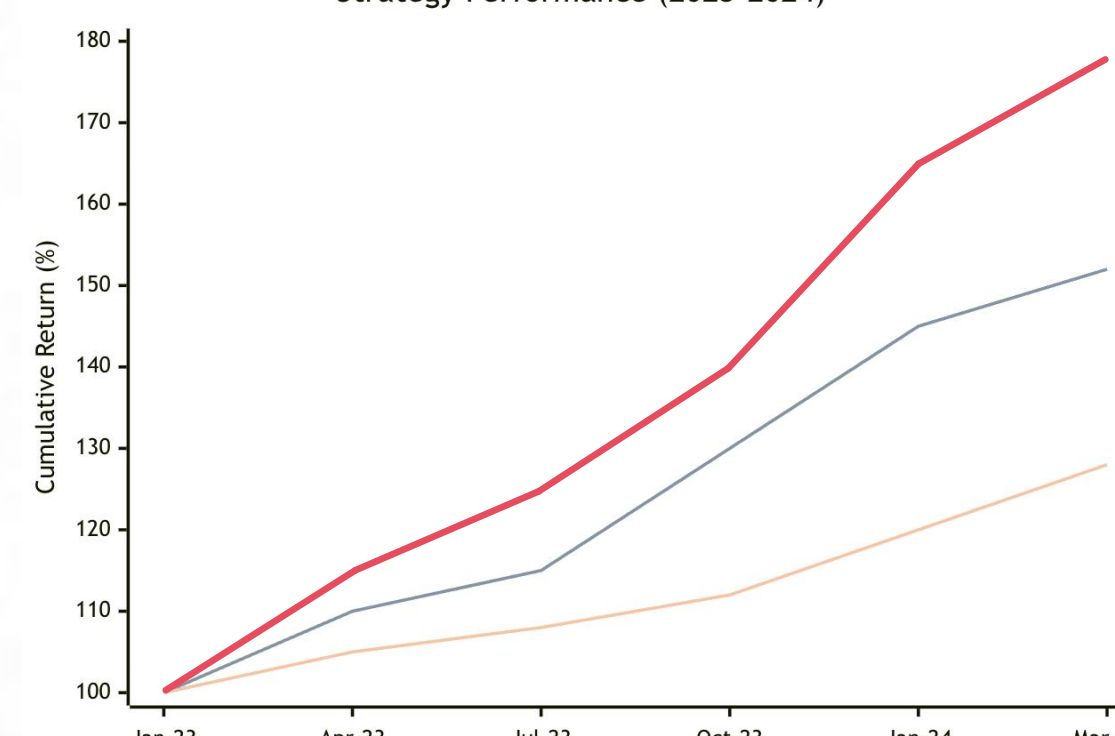
FlowPrint vs. Momentum Strategy: 49% higher return, 22% smaller maximum drawdown

Optimal Cash Flow Weight: 35%-40% (best risk-adjusted return)

Sample Ranking (2024-12-31): Technology > New Energy > Financials > Healthcare > Consumer



Strategy Performance (2023-2024)



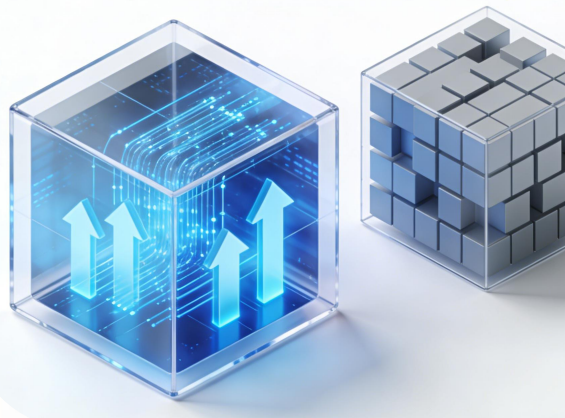
Practical Implications

Individual Investors: Systematic sector allocation tool

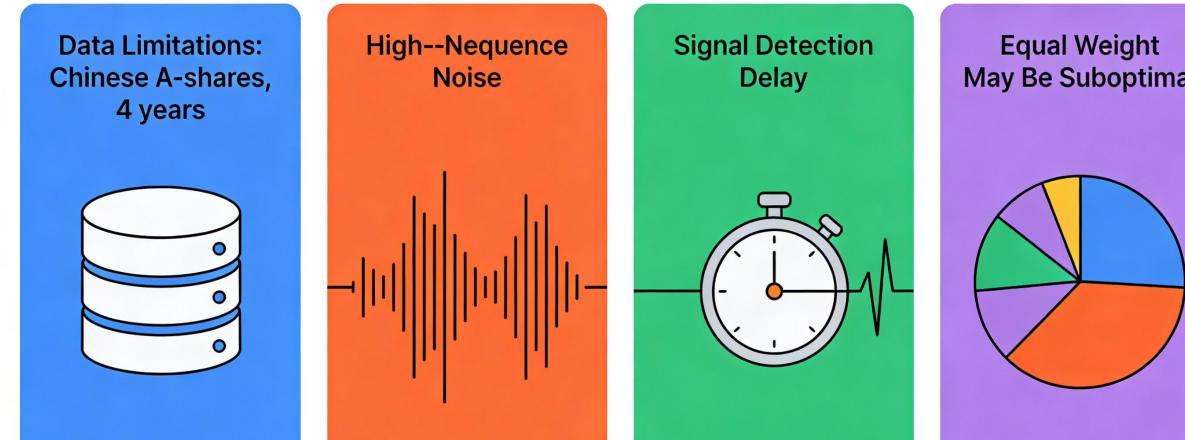
Institutions: Customizable complementary quantitative strategy

Advantages: Fully automated (time-saving), transparent & interpretable (trust-building), modular design (flexible)

Fund Flow Forecasting ETF vs Traditional ETF



Problems



References

- [1] Fama, E. F., & French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3-56.
- [2] Moskowitz, T. J., Ooi, Y. H., & Pedersen, L. H. (2012). Time series momentum. *Journal of Financial Economics*, 104(2):228-250.
- [3] Ben-David, I., Franzoni, F., & Moussawi, R. (2018). Do ETFs increase volatility? *The Journal of Finance*, 73(6):2471-2535.